



Chapter 15

An Algebraic Equivalence between Generalized Fixed Effects and a Generalized Mundlak Regression with Applications to Heterogeneous Trends and Difference-in-Differences

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Abstract Using a modified version of the Frisch-Waugh-Lovell partialling out result, I show how the standard Mundlak regression for panel data can be extended to deliver ‘generalized fixed effects’ estimators, where coefficients on a subset of explanatory variables vary by cross-sectional unit. The characterization shows that only fitted values from individual specific regressions need to be controlled for – not the large set of interactions between unit-specific dummy variables and explanatory variables. In the special case of a single additive heterogeneity, the fitted values are simply the unit-specific time averages of the explanatory variables. The general results have practical applications, particularly when testing the standard fixed effects estimator against heterogeneous trend models. I derive a new, fully robust test that can be viewed as a regression-based Hausman test for choosing the level of heterogeneity. As an example, I reconsider some recently proposed difference-in-differences methods with staggered interventions and heterogeneous treatment effects when the parallel trends assumption is possibly violated.

15.1 Introduction

The classic paper by Balestra and Nerlove (1966), showing how dynamic economic relationships with unobserved heterogeneity can be estimated using panel data, is now almost 60 years old. Now, the vast majority of intervention and policy analysis studies rely heavily on panel data structures, and the wealth of methods that have been developed since the early work by Mundlak (1961), Balestra and Nerlove (1966), and others.

In linear panel data analysis, the algebraic equivalence of the fixed effects estimates (on time-varying covariates) and estimating the Mundlak (1978) equation – which adds the unit-specific time averages of time-varying explanatory variables – by random effects has many applications. In Wooldridge (2019), I showed not only does the equivalence carry over to unbalanced panels, but also that pooled ordinary least squares (POLS) estimation of the Mundlak equation is identical to the random effects generalized least squares (GLS) estimator. I referred to the POLS version of the estimation as the ‘Mundlak regression’.

An important application of the algebraic equivalence between the Mundlak regression and the fixed effects estimator is obtaining a fully robust approach to choosing between the random effects

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and fixed effects estimators: a robust, regression-based version of the Hausman (1978) test. It also suggests that the Mundlak device can be useful for modeling the relationship between heterogeneity and covariates in nonlinear panel data models where no consistent fixed effects estimators (with the number of time periods fixed in the asymptotic analysis) are available. See, for example, (Wooldridge, 2010, Chapters 13, 15, 16, and 17).

In Wooldridge (2024), I extended the equivalence between fixed effects and the Mundlak regression to explicitly include time effects. One way to characterize the so-called ‘two-way fixed effects’ (TWFE) estimator is that one includes dummy variables for each time period (less one) and each unit. The ‘two-way Mundlak regression’ adds an intercept and the cross-sectional averages of the explanatory variables for each time period t to the POLS estimation. Baltagi (2023) shows how this equivalence leads to several insights concerning estimation of average treatment effects in difference-in-differences settings with staggered interventions. Baltagi (2023) showed that a GLS estimator that extends the usual one-way random effects estimator is equivalent to the pooled OLS estimator.

In many applications, one wants to allow for additional heterogeneity, most commonly via unit-specific trends. Even in microeconomic applications with a handful of time periods it can be important to allow units to have different trends in the outcome variable and the covariates. In a difference-in-differences setting, the outcome in the untreated state might be trending differently for control units and treated units, and even among units treated for the first time in different time periods. Such a situation would arise if the intervention decision is not based merely on levels difference before the intervention but also on pre-existing differences in trends.

Many researchers use an extended fixed effects estimator that includes unit-specific dummy variables along with those same dummy variables interacted with a linear time trend. An application of the Frisch-Waugh-Lovell (FWL) Theorem shows that including the $2N$ regressors (where N is the cross-sectional sample size) is equivalent to using unit-specific detrending of the covariates (and, optionally, the response variable). See, for example, Wooldridge (2010), Section 11.7.2.

In this paper, I extend the Mundlak regression to allow for a general model where a subset of variables has unit-specific coefficients. As discussed by Chamberlain (1992), the fixed coefficients can be estimated by generalizing the within transformation used by the simplest fixed effects estimator. Here, I show that the partialling out approach proposed by Chamberlain (1992) – see also Wooldridge (2010), Section 11.7.2 – is equivalent to adding unit-specific fitted values. The proof of equivalence relies on a general result on the equivalence between adding fitted values and using FWL partialling out. Though the equivalence, which I establish in Section 15.2, is straightforward matrix algebra, it has applications to allowing lots of heterogeneity in panel data settings.

In Section 15.3, I show, in the context of panel data, how one can obtain Chamberlain’s ‘generalized fixed effects’ (GFE) estimator using a ‘generalized Mundlak regression’ (GMR). The GMR consists of adding the fitted values from first-stage unit-specific regressions, where the explanatory variables with fixed coefficients are regressed on the explanatory variables with heterogeneous coefficients. The first step results in functions of the explanatory variables for each unit i . Then, the outcome variable is regression on the explanatory variables with fixed coefficients and the unit-specific fitted values.

In Section 15.4 I consider the important special case of heterogeneous trends – which can themselves be general. The key is that the variables with heterogeneous coefficients change only across time, and not by unit. Again, this generalizes the usual Mundlak regression by putting in fitted values that are linear combinations of the time trend for each unit i . The result is a relatively short regression that nevertheless allows unrestricted heterogeneity on a subset of the regressors. I also show how aggregate time variables with fixed coefficients – with the leading case being a full set of time period dummies – can be accommodated. Also, for the purposes of testing whether enough heterogeneity has been allowed, I show what changes when time-constant covariates are interacted with general time trends. I then apply the characterizations to specification testing in Section 15.5. It is straightforward to use a variable addition test to determine whether less heterogeneity is sufficient against the alternative that more is needed in the context of heterogeneous trends models. The leading case is when the usual two-way fixed effects (TWFE) estimator has been computed, but one may